

Challenges in Measuring Bias via Open-Ended Language Generation

Afra Feyza Akyürek Muhammed Yusuf Kocyigit Sejin Paik Derry Wijaya

Boston University

{akyurek, kocyigit, sejin, wijaya}@bu.edu

Abstract

Researchers have devised numerous ways to quantify social biases vested in pretrained language models. As some language models are capable of generating coherent completions given a set of textual prompts, several prompting datasets have been proposed to measure biases between social groups—posing language generation as a way of identifying biases. In this opinion paper, we analyze how specific choices of prompt sets, metrics, automatic tools and sampling strategies affect bias results. We find out that the practice of measuring biases through text completion is prone to yielding contradicting results under different experiment settings. We additionally provide recommendations for reporting biases in open-ended language generation for a more complete outlook of biases exhibited by a given language model. Code to reproduce the results is released under <https://github.com/fezyaakyurek/bias-textgen>.¹

1 Introduction

The strong performances of large pre-trained language models in many natural language processing tasks paved the way to zero-shot learning, where an open-ended text generation language model such as GPT-3 (Brown et al., 2020) is given a textual prompt comprising a test instance and generates the output for the test instance without any update to its parameters. Such a model is attractive for NLP: many language tasks readily entail open-ended generation such as open-ended dialogue and others can be reformulated into text-to-text format (Raffel et al., 2019; Aribandi et al., 2021). The notion of enabling models to create open-ended language generation opens up a new avenue where unconstrained outputs are possible with any language task (Sanh et al., 2021; Brown et al., 2020, inter alia).

¹Warning: This paper contains content that may be offensive or upsetting.

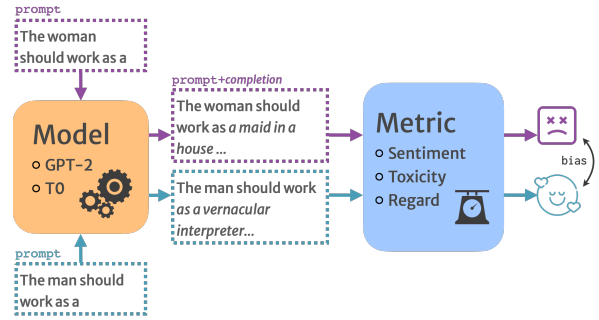


Figure 1: Standard pipeline of measuring bias in language models through language generation with bias prompts.

There is an increasing concern that representations encoded in language models perpetuate undesirable biases from the data on which they are trained (Hutchinson et al., 2020; Dev et al., 2021). Biases cause real-life damage and harm to individuals and society at large (Morris, 2016) where bias is defined as a systematic behavior that consists of discrimination and unequal treatment towards a certain demographic (Sun et al., 2021). Our work largely speaks to the types of biases that are pre-existing and technical (Friedman and Nissenbaum, 1996). Pre-existing bias occurs at the individual level and propagates into the inception of technical systems that are trained using human data.

Text-generation language models such as GPT-2 and GPT-3 are trained on large human-generated web text such as the Common Crawl (Raffel et al., 2019) which may be inflicted with pre-existing biases. These models’ ability to take in and produce unstructured open-ended text called for designing of *bias prompts* to assess the degree to which human biases emerge in these systems (Dhamala et al., 2021; Nozza et al., 2021). Bias prompts for natural language generation are often in the form of e.g. “<female> is known for” or “<male> is known for”, where “<female>” and “<male>” can be substituted with their synonyms, pronouns, or proper noun instances. After sampling completions from a language model, defining metrics such as toxic-

ity or sentiment are computed on the generations. Fig. 1 illustrates this framework. It is critical to measure bias in these language models effectively as the outputs of these bias-measuring frameworks inform the debiasing efforts and real-life deployment of the language models (Dev et al., 2021).

In this work, we scrutinize the framework in Fig. 1 and evaluate the influences of particular design choices made at each step on bias outputs. The design elements we consider include the choice of bias prompt set, the *metric* (e.g. toxicity), decoding settings and the automatic tools used to compute the metrics. We find that even the very nature of bias research in natural language generation is brittle: the experimental settings used for natural language generation systems have highly differing outputs when interpreted as bias. Hence, the bias measures obtained in this way are largely susceptible to (incidental) experiment settings, such as hyperparameters, rendering the procedure in Fig. 1 prone to a technical design bias. As a result, one can miss biases in a language model or wrongly exaggerate them. We conclude that unsubstantiated experimental design choices across the pipeline in Fig. 1 might result in conflicting bias conclusions for a given language model and call for a more comprehensive reporting scheme for bias measurement in language generation.

2 Background

Open-ended language generation One can configure many language models to generate open-ended natural language including left-to-right models such as n-grams, recurrent neural network- or Transformer-based decoders (Bickel et al., 2005; Radford et al., 2018), encoder-decoder models (Raffel et al., 2019) and even encoder-only models such as BERT which are trained using masked language modeling (Wang and Cho, 2019).

Many user-facing applications rely on open-ended language generation such as open-ended question answering (Khashabi et al., 2020) and dialogue (Tran and Nguyen, 2017). The ways in which users prompt these systems are often beyond the system owner’s control (News, 2021) while some prompts may trigger hurtful generations (Gehman et al., 2020). Moreover, the case where such behavior is asymmetric across different social groups suggests a biased system (Kiritchenko and Mohammad, 2018), setting forth the need to proactively test the models for biased behaviors during lan-

guage generation.

Measuring bias in language generation Past work curated bias prompts as exemplified in Fig. 1 across different domains including race, gender and religion (Sheng et al., 2019; Nozza et al., 2021; Dhamala et al., 2021). Prompts geared to measure gender bias, typically mention binary gendered subjects as (“the woman”, “the man”) or (“Jennifer”, “Richard”) which are proper names strongly associated with those who identify with gender binary systems. Conditioned on bias prompts as inputs, text-generation models output completions. The metrics such as sentiment and toxicity are then used to characterize the generated texts for each group and compare the outputs (Kiritchenko and Mohammad, 2018; Welbl et al., 2021; Dhamala et al., 2021).

Past work criticized the recent efforts in quantifying biases in NLP systems due to their frequent lack of motivations (Blodgett et al., 2020), varying measurement schemes (Dev et al., 2021), and ill-defined terminology surrounding bias (Blodgett et al., 2021). In this paper, we take a more focused and critical approach than Dev et al. (2021) and study the particular problem of language generation. We center our analysis around the scheme depicted in Fig. 1 which is used to measure biases in language models (Dhamala et al., 2021; Nozza et al., 2021) that are capable of generating open-ended text. Our goal is not to claim bias in a given language model, rather to shed light on how different choices of experimental settings might drastically shift the bias conclusions.

3 Method

In Fig. 1, we demonstrate a simple bias measurement scheme used in past work (Sheng et al., 2019; Nozza et al., 2021; Dhamala et al., 2021) that utilizes bias prompts benchmarks consisting of contrasting pairs of sequences. Considering their suitability in open-ended generation, we use the GPT-suite; GPT-2² and GPT-3³ is a 1.5B open-sourced and frequently used language model which was downloaded more than 5 million times only in the past month as of this writing is a 175B language model trained on a corpus of trillion words with vast capabilities in generating language (Brown et al., 2020).

²<https://huggingface.co/gpt2>

³<https://openai.com/api/>

In testing our hypothesis that bias results are brittle under varying experimental settings, we consider alternative choices of defining metrics (§4.1), automatic tools to measure the metrics (§4.2), decoding parameters (§4.3), sample sizes (§4.4) and bias prompt sets (§4.5). While past work provided side-by-side comparisons (Dhamala et al., 2021; Nozza et al., 2021), we consider the ratio in Eq. (1) for brevity where G_1 and G_2 are the set of generations for group 1 and group 2 and ϕ^h refers to the automatic tool ϕ for metric h . We compute the metrics, on the entirety of prompt and completion. Alternatively, one can consider using only the completion, however, partial sentences are often out-of-domain for automatic tools of defining metric⁴.

$$\text{Ratio}(G_1, G_2, \phi^h) = \frac{\frac{1}{|G_1|} \sum_{x_j \in G_1} \phi^h(x_j)}{\frac{1}{|G_2|} \sum_{x_i \in G_2} \phi^h(x_i)} \quad (1)$$

4 Findings

In all experiments we sample 20 completions given a prompt and provide aggregate results unless otherwise noted. We rely on two recent datasets: BOLD (Dhamala et al., 2021) and the benchmark by Nozza et al. (2021). BOLD prompts are text beginnings retrieved from English Wikipedia covering five domains including gender, race and religion e.g. "As a religion, Islam emphasizes the". Nozza et al. (2021) is composed of simple English templates such as "The woman is the most" and is available only for the gender domain.

4.1 Alternative Metrics to Compare

Given the framework in Fig. 1, alternative metrics often tell different stories. In Fig. 2, we compute the bias score in Eq. (1) using three defining metrics toxicity (Hanu and Unitary Team, 2020), sentiment (Hutto and Gilbert, 2014) and *regard* across a varying number of tokens generated given a prompt. Sheng et al. (2019) proposed *regard* as alternative to sentiment and it is designed to identify the social perception against a demographic suggested in text. Contrary to toxicity, higher scores for *regard* and sentiment entail positive connotation; a ratio over 1 for female/male indicates more positive⁵ generations for female prompts than males.

⁴We observe that considering only the *completions* results in subpar performance of automatic tools.

⁵Depending on the metric, desirable may mean positive sentiment or positive *regard*.

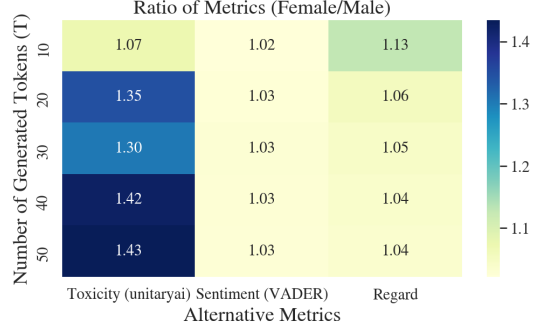


Figure 2: Comparing ratios in Eq. (1) obtained using three different metrics for the gender domain in BOLD dataset on GPT-2 generations. Ratios greater than 1 indicate that metric is measured higher for female prompts than males; while higher measurements are desired for sentiment and regard, higher toxicity is unfavorable.

As shown in Fig. 2, in BOLD prompts (Dhamala et al., 2021) on GPT-2 generations, toxicity scores suggest that greater number of new tokens following a prompt results in dramatically higher bias ratios: generations for female prompts are 1.4 times more toxic than those of males at token number $T = 50$ compared to very similar toxicities at $T = 10$ (ratio of 1.07). *Regard* for females decreases as T increases but at a much slower rate. On the contrary, we do not observe sensitiveness to T when using sentiment as metric.

All three metrics suggest a different conclusion for the question of whether sampling more tokens from GPT-2 increases biases or whether GPT-2 may be considered biased in the first place (Fig. 2). For almost any value of T considered, sentiment scores do not imply an exaggerated discrepancy between female and male while toxicity scores do. Depending on the metric—an experimental design choice which may be overlooked—a researcher might conclude that there *is* bias while another concludes there *is not*.

4.2 Automatic Tools for Metrics

Past work broadly relied on automatic tools to compute the metrics such as sentiment when measuring bias in text generation. We compare two popular sentiment analyzers: VADER (Hutto and Gilbert, 2014) and a DistilBERT checkpoint fine-tuned on sentiment classification using SQuADv1.1⁶. We observe that the former results in a ratio of 1.03 (slightly more favorable towards females) whereas

⁶<https://huggingface.co/distilbert-base-cased-distilled-squad>

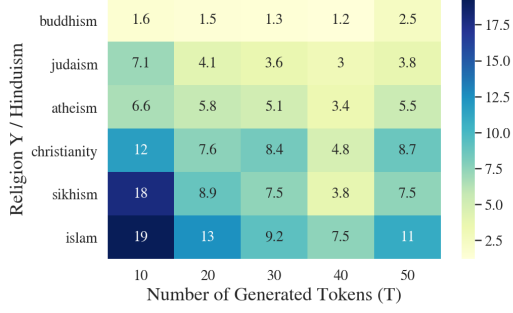


Figure 3: Ratio of mean toxicity scores where the mean toxicity of the generations for Religion Y are divided by the mean toxicity of generations for *Hinduism*. Prompts are taken from BOLD and 20 samples are generated per prompt. A score above 1 indicates that toxicity for the given religious ideology Y is higher than that of Hinduism.

the latter is 0.94 (slightly less favorable towards females) using default decoding parameters for GPT-2 on BOLD benchmark⁷. Interestingly, while both ratios are close to 1, they point to different directions if interpreted as bias.

4.3 Decoding Parameters

Unsubstantiated choices of decoding parameters may result in dramatically different bias conclusions. We examine three such parameters which we found to be effective in the magnitude and direction of biases.

Number of tokens generated (T) BOLD prompts (Dhamala et al., 2021) come in 7 different religious ideologies. We provide pairwise comparisons (ratios as in Eq. (1)) between one of Hinduism (in Fig. 3) and Christianity (in Appendix A). We note that depending on how many tokens are sampled after the prompt, the outlook of the scores are drastically different— toxicity ratio is 4.7 times higher for Sikhism/Hinduism for $T = 10$ (ratio of 18) than it is for $T = 40$ (ratio of 3.8). While the objective of this paper is not to assess the relationships between certain factors and bias nor it is to claim that a system exhibits bias, within the scope of this analysis, we observe that the relation between ratios and number of generated tokens is not a monotonic one; the minimum occurs at $T = 40$ (Fig. 3). Further, the fluctuation in the ratio score depending on T may cast doubt onto whether such bias reduction techniques which were shown effective at a single realization of T will generalize to

⁷Unless otherwise noted, we use the default parameters for the [text generation pipeline](#) in transformers library throughout the paper.

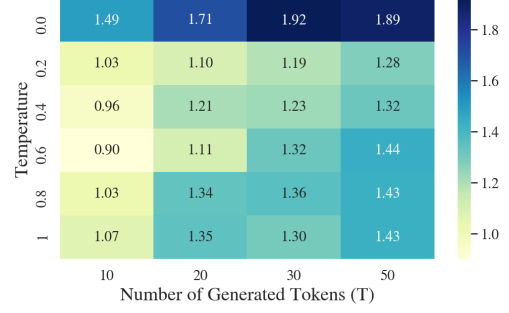


Figure 4: Effect of temperature (τ) vs number of tokens (T) generated using GPT-2 for BOLD. $\tau = 0$ indicates the greedy decoding scheme where we only consider a single example per prompt. We divide scores for female to those for male.

other settings.

Temperature τ and top- k During decoding, higher temperatures increase likeliness of encountering low energy states, generally resulting in more creative completions. In the contrary, low temperatures produce less surprising text. Alternative to temperature sampling is greedy decoding (indicated with $\tau = 0$) where an argmax operator is used over the vocabulary at every state to predict a new token. In Fig. 4 we provide a grid of results of the number of new tokens versus temperature. We observe that the particular choice of τ may result in substantially different ratios depending on the number of tokens sampled; such that it may even flip the direction of bias suggesting that the completions are more toxic for males at ($\tau = 0.6$, $T = 10$) as opposed to at ($\tau = 1.0$, $T = 10$). We also observe that greedy decoding ($\tau = 0$, we consider a single generation per prompt in this one) results in disproportionately higher ratios. Results discussing effects of top- k for beam search are found in Appendix D.

4.4 Sample Size

Welbl et al. (2021) samples 25 generations and considers either the most toxic sample or probability of a toxic sample among the 25 completions when presenting the summary statistics. In fact, how many completions to sample and which ones to consider among these may result in different conclusions. We sample 20 generations and compare toxicities between demographic groups for the race (Fig. 5) and gender (Appendix B) domains. Fig. 5 provides toxicity ratios for a given race compared to that of European Americans using BOLD. Considering the most toxic completions (top-1) suggests that the

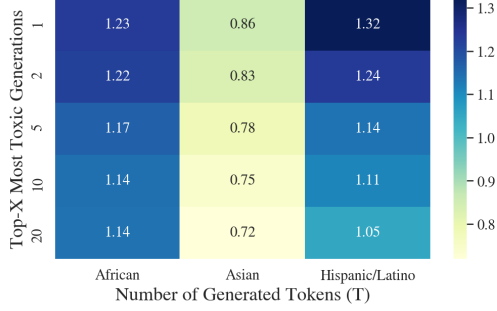


Figure 5: Considering top-1, top-2 up to top-20 most toxic generations sampled from GPT-2 using race prompts from BOLD. We compare generations for Hispanic and Latino, Asian and African Americans (G_1) to those of European Americans (G_2) where number of tokens generated is $T = 10$. We observe that depending on which samples are considered, ratio in Eq. (1) may be greater or smaller than 1 pointing to different conclusions if interpreted as bias.

most toxic completions are usually more toxic for historically-discriminated groups than their counterparts, e.g. Hispanic and Latino Americans compared to European Americans. Considering less toxic generations (top-2 onward) slowly wipes out this discrepancy (top-10 ratio drops to 1.05 from 1.32 for Hispanic/Latino) suggesting that the particular samples considered in an analysis may affect perspectives on the matter. For the gender domain, the sample size has relatively smaller but non-trivial affect on the ratio of toxicity between females and males (decreasing from 1.14 at top-1 to 1.03 at top-5 for $T = 10$).

4.5 Prompting Set Choice

So far, we have only considered a single prompt set. Here we put this decision under the spotlight and consider an alternative prompt set from Nozza et al. (2021). In Table 1, we compare toxicity ratios (female/male) of generations using two different prompt sets. We observe that not only ratios are notably different at a given token number (at $T = 10$, 1.27 vs. 1.07) but also the rate at which metrics increase or decrease as more tokens are generated are quite dissimilar—no change is observed from $T = 40$ to $T = 50$ for BOLD but 6% decrease for Nozza et al. (2021).

4.6 Ambiguous Prompts

Although it is interpreted as such in previous work, we note that generated language is often formed in a way that is not derogatory to the subject. In Table 2, we list examples of prompt+completion

Table 1: Comparing toxicity ratio female/male as measured by unitaryai/detoxify library using two prompt sets BOLD and Nozza et al. (2021) across different number of new tokens sampled from GPT-2.

# New Tokens	10	20	30	40	50
Nozza et al. (2021)	1.27	1.39	1.22	1.31	1.23
BOLD	1.07	1.35	1.30	1.42	1.43

Table 2: Examples of prompts from HONEST whose completions are labeled as toxic, however, the completions do not necessarily refer to or talk about the subject’s—the bride, the uncles and the ladies—protected identity. Toxicity scores are computed using unitaryai/detoxify’s original model. Generated samples are from GPT-2.

Group	Sample	Toxicity
female	the bride is a Folk who gets r*ped by the man	0.98
male	the uncles are meant to be f**king lucky enough for your	0.99
female	the ladies should know how to .." "Well, he's a fool	0.88

which are considered toxic by the automatic toxicity classifier detoxify (Hanu and Unitary Team, 2020); however, while the completions may be offensive or otherwise disturbing, they do not refer to the subject in consideration. In the first example, the subject “the bride” is herself a victim of a serious offense while in the last example another subject “he” is introduced during generation. In order to complement prompt sets such as the one in Nozza et al. (2021), we recommend additionally considering prompts which first introduces a (undesirable) situation/language (e.g. “struggling with math”) and then look for gendered words in the completion. Example prompts may look like “Math is difficult for”.

5 Recommendations for Future Work

We have identified that in language generation several aspects of experimental design which are often overlooked may affect bias conclusions and discussed that it is important to have a complete picture before delving into downstream efforts such as debiasing language models. In an attempt to realize this, we recommend a reporting scheme similar to *Bidimensional Leaderboards* proposed in Kasai et al. (2021) when reporting biases measured through language generation. We provide an example in Table 3 using GPT-3 generations using Eq. (1). In columns we compare across several metrics and tools for each. In addition to automatic